

DATABASE MANAGEMENT SYSTEM PROJECT

Health Insurance Policy & Claims Management System

Project 53 | Review 1

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Problem Identification



Fragmented operations

Policies, premiums, dependants, hospitals, claims, documents, assessments, settlements and appeals are managed across many disconnected spreadsheets and tools.



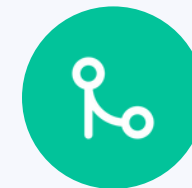
Tracking difficulty

Manual, disconnected processing causes duplicate claims and makes it hard to see a claim's current status, owner or history.



Need for a single source of truth

A centralized relational database is needed to keep policy and claim data consistent and enforce referential integrity.



End-to-end integration

The proposed system connects policy administration with the full claim workflow, from submission through settlement and appeal.

Objectives



Design a structured relational database for health insurance policy and claim management end-to-end.



Reduce duplicate and inconsistent records through keys, constraints and defined relationships.



Maintain customer, dependant, policy, premium, hospital and claim data in one integrated system.



Track claims end-to-end: submission, verification, assessment, decision and settlement.



Support appeals on rejected claims and generate reports for management review.



Prepare a normalized (3NF) design that can be implemented in MySQL, PostgreSQL or Oracle.

Scope of the System

1

Policy management — Create, renew and update policies linked to a customer and plan.

3

Customer & dependant records — Maintain profiles for policyholders and their covered dependants.

5

Plan & premium management — Define coverage plans and generate recurring premium dues.

7

Payment records — Log premium payments and track payment mode and status.

9

Hospital information — Maintain hospital and service-provider contact details.

2

Claim submission & verification — Capture new claims and verify eligibility against the active policy.

4

Treatment & document records — Store treatment details and supporting documents per claim.

6

Assessment & decisions — Evaluate claimed amounts and record approve / reject decisions.

8

Settlement & appeals — Process approved payouts and manage appeals on rejected claims.

10

Reporting & database operations — Support queries and reports across the full data model.

Scope of the System

Role	Responsibilities
Insurance Administrator	Manages plans, policies, customers and dependants; the primary user for onboarding and policy issuance.
Claims Officer	Handles claim submission, verification and processing; first point of contact once a claim is filed.
Assessment Officer	Records assessed amounts and claim decisions; determines what portion of a claim is approved.
Accounts / Finance User	Manages premiums, payments and settlements; reconciles amounts due against amounts received.
Hospital / Service User	Provides hospital and treatment-related claim information, including treatment cost and diagnosis detail.
Management	Views reports such as active policies, premium dues, pending claims, approval rate and settlements.

Thank You
